Shuo Zhang

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Education

Duke University – Durham, NC 2025–2030 (expected)
PhD in Computer Science
Advisors: Prof. Sudeepa Roy & Prof. Jun Yang

Columbia University – New York, NY Dec 2024
MS in Computer Science (GPA: 4.0/4.0)

Boston University – Boston, MA May 2023
BA in Computer Science & Economics (GPA: 3.83/4.00, magna cum laude)

Research Interests

AI for Data Management · Database Usability · Query Validation · Formal Verification

Research Experience

Research Assistant, Duke Database Group – Durham, NC Aug 2025 – present

- **Reference-Free SQL Validation:** Designed and implemented a validator-aware query-decomposition optimizer using semantics-preserving plan rewrites, dynamic programming, and A* search to balance modeled validation reliability against the number of required judgments.
- On 305 candidate SQL queries from BEAVER and Spider 2.0, improved LLM error-detection recall by 7.0–16.9 percentage points over whole-query validation at over 90% precision; a user study showed human error-detection recall rising from 36% to 91% with decomposition.
- **Whiel: Certified Relational Program Verification:** Implemented an early verification pipeline in Lean, translating relational algebra into first-order formulas in TPTP format and invoking Vampire to check loop-invariant verification conditions.
- Led the design of an 86-task benchmark for relational program verification and built evaluation baselines; the full system solved 85% of tasks versus 23% for direct LLM-generated Lean proofs, at about one-quarter of the baseline's token cost.

Graduate Researcher, Columbia University Database Group – New York, NY May 2024 – Jan 2025

- **LLM-Based Data Cleaning:** Co-developed Cocoon, which pairs LLM reasoning with statistical error detection and beats HoloClean and Raha+Baran on 4 of 5 benchmarks by up to 0.27 F1.
- **Text-to-SQL (NL2SQL) Analysis:** Identified data-quality issues affecting up to 37% of ground-truth answers in case studies on the BIRD benchmark.
- **Query Execution:** Prototyped a hybrid execution strategy that dynamically selects between hash joins and sparse/dense matrix multiplication for join-aggregate queries—up to 6x faster than DuckDB in early tests.

Graduate Researcher, Columbia Internet Real-Time Lab – New York, NY Sept 2023 – Jan 2025

- **IoT Policy Framework:** Designed the Identity-Independent Policy framework to express up to 99% of policies in real-world IFTTT datasets; implemented a Python policy-server prototype.

Undergraduate Researcher, Boston University DiSC Lab – Boston, MA June 2022 – May 2023

- **SSD Graph Processing (CAVE, SIGMOD 2024):** Co-developed CAVE, a C++ graph processing engine that exploits SSD parallelism to run up to 984x faster than GraphChi; implemented its parallel BFS, DFS, WCC, PageRank, and random walk.
- Developed a blocked variant of CAVE that avoids repeated block reads and performs better with small caches—up to 11x faster than the non-blocked version.

Publications & Manuscripts

Optimizing Query Decomposition for Reliable Validation 2026
Shuo Zhang, Sudeepa Roy, Jun Yang
Under review at VLDB 2027

Whiel: A Neurosymbolic Platform for Certified Relational Program Verification 2026
Jesse Comer*, Fangzhu Shen*, ***Shuo Zhang****, Mayur Naik, Sudeepa Roy, Val Tannen
arXiv:2611.05612; Under review at VLDB 2027

Advancing IoT System Dependability: A Deep Dive into Management and Operation Plane Separation 2025
Luoyao Hao, ***Shuo Zhang***, Henning Schulzrinne
IEEE World Forum on Internet of Things (WF-IoT)

Data Cleaning Using Large Language Models 2025
Shuo Zhang, Zezhou Huang, Eugene Wu
ICDE Workshop on Data-AI Systems (DAIS), Short Paper

Data-Centric Text-to-SQL with Large Language Models 2024
Zezhou Huang, ***Shuo Zhang***, Kechen Liu, Eugene Wu
NeurIPS Workshop on Table Representation Learning (TRL)

Policy Enforcement for IoT: Complexities and Emerging Solutions 2024
Shuo Zhang, Luoyao Hao, Henning Schulzrinne
IEEE International Performance, Computing, and Communications Conference (IPCCC), Poster

CAVE: Concurrency-Aware Graph Processing on SSDs 2024
Tarikul Islam Papon, Taishan Chen, ***Shuo Zhang***, Manos Athanassoulis
Proceedings of the ACM on Management of Data (SIGMOD)

Industry Experience

Summer Intern, VisionX LLC – San Jose, CA June 2022 – Aug 2022

- Fine-tuned a YOLO-based PyTorch detector for crop pests and diseases, achieving a 1.8x processing speedup over the previous product.

Teaching Experience

- Teaching Assistant, CompSci 316: Introduction to Database Systems, Duke University, Fall 2026
- Teaching Assistant, CompSci 516: Database Systems, Duke University, Spring 2026

Skills

Languages: Python, C++, SQL, Java, Rust, Datalog, R, JavaScript
Databases: PostgreSQL, DuckDB, SQLite, MongoDB
Formal Verification: Lean 4, Vampire
Tools & Libraries: Spark, Calcite, SQLGlot, egglog, Git, Docker, PyTorch, scikit-learn, pandas